CAMBRIDGE
International Education

Cambridge International AS & A Level

MATHEMATICS **9709/11**

Paper 1 Pure Mathematics 1 **October/November 2024**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **15** printed pages.

© Cambridge University Press & Assessment 2024

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© Cambridge University Press & Assessment 2024

Mathematics Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

© Cambridge University Press & Assessment 2024

PUBLISHED

Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

© Cambridge University Press & Assessment 2024

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
1	Identify correct term and obtain $6(kx)^2 \cdot \left(\frac{2}{x}\right)^2$	**M1**	Needs numerical coefficient or $\frac{4!}{2!2!}$, not ${}^4C_2$.
	Equate to 150 and obtain $k = \frac{5}{2}$	**A1**	Ignore $-\frac{5}{2}$
	Identify correct term $4(kx)^3 \cdot \left(\frac{2}{x}\right)$ with their value of k	**M1**	Needs numerical coefficient or $\frac{4!}{3!1!}$.
	Obtain coefficient 125	**A1**	Accept $125x^2$ as final answer.
		4	

Question	Answer	Marks	Guidance
2	Differentiate to obtain $2x + ax^{-2}$ or equivalent	**B1**	
	Equate first derivative to zero, substitute $x = -3$ and attempt value of a	**M1**	Must be an attempt at differentiation.
	Obtain $a = 54$	**A1**	
	Obtain $b = 27$	**A1**	
		4	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
3	Use correct sector area formula	**M1**	
	Obtain $\frac{1}{2}\times 15^2\times\frac{2}{5}\pi - \frac{1}{2}\times x^2\times\frac{2}{5}\pi = \frac{209}{5}\pi$ or equivalent	**A1**	
	Obtain $[x]=4$	**A1**	AWRT 4.00.
	Use correct arc length formula twice	**M1**	
	Obtain $22+\frac{38}{5}\pi$	**A1**	OE. Must be in terms of π. Like terms must be collected. Not from a rounded value of x.
		5	

Question	Answer	Marks	Guidance
4	Substitute for y (or x) in first equation and simplify	***M1**	All terms to one side and brackets expanded.
	Obtain $10x^2+3kx-40$ [= 0] (or $10y^2+11ky+k^2-360[=0]$)	**A1**	
	Attempt b^2-4ac for 3-term quadratic involving k	**DM1**	Not in quadratic formula unless b^2-4ac is isolated.
	Obtain $9k^2+1600$ (or $81k^2+14400$)	**A1**	
	$9k^2+1600>0$	**A1 FT**	FT for $ak^2+b>0$ with $a, b>0$.
		5	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
5(a)	Attempt correct process for solving 3-term quadratic equation in $\sqrt{x}$	**M1**	Accept $8y^2 - 6y - 9 \rightarrow (2y - 3)(4y + 3)$, if $y = \sqrt{x}$ specified.
	Obtain at least $2\sqrt{x} - 3 = 0$ or equivalent	**A1**	Ignore $4\sqrt{x} + 3 = 0$. **SC B1** for $\sqrt{x} = \frac{3}{2}$ with no method shown for solving the 3-term quadratic.
	Conclude $x = \frac{9}{4}$ ignore $\frac{9}{16}$	**A1**	**SC B1** if no method shown for solving the 3-term quadratic.
	Alternative Method for Q5(a)		
	$3\sqrt{x} = 4x - \frac{9}{2} \rightarrow 16x^2 - 45x + \frac{81}{4}$ o.e and attempt correct process to solve	**M1**	
	Obtain $x = \frac{9}{4}$ or $\frac{9}{16}$	**A1**	**SC B1** if no method shown for solving the 3-term quadratic.
	$x = \frac{9}{4}$ ignore $\frac{9}{16}$	**A1**	**SC B1** if no method shown for solving the 3-term quadratic.
		3	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
5(b)	Integrate to obtain form $k_1x^2 + k_2x^{\frac{3}{2}} + k_3x$ where $k_1k_2k_3 \neq 0$	**M1**	
	Obtain correct $2x^2 - 2x^{\frac{3}{2}} + x$ or equivalent	**A1**	Allow unsimplified.
	Substitute $x = 4$ and $y = 11$ to attempt value of c	**M1**	Dependent on at least 2 correct terms involving x.
	Obtain $y = 2x^2 - 2x^{\frac{3}{2}} + x - 9$	**A1**	Must be simplified. Allow 'f(x) ='. Allow y missing if y appears previously.
		4	

Question	Answer	Marks	Guidance
6(a)	State or imply centre of C_1 is $(-3, 5)$	**B1**	
	State or imply centre of C_2 is $(9, -4)$	**B1**	
	Attempt correct process for finding distance between centres	**M1**	
	Obtain 15	**A1**	
		4	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
6(b)	$R = 4$ and $R = 8$	**B1**	
	Obtain least or greatest distance	**B1 FT**	'15' – R_1 – R_2 or '15' + R_1 + R_2.
	Obtain 3 and 27	**B1 FT**	'15' – R_1 – R_2 and '15' + R_1 + R_2.
		3	

Question	Answer	Marks	Guidance
7(a)	Differentiate to obtain form $k_1(2x+1)^{-\frac{4}{3}}$	**M1**	
	Obtain correct $-8(2x+1)^{-\frac{4}{3}}$ or unsimplified equivalent	**A1**	
	Attempt equation of tangent at $\left(\frac{7}{2}, 6\right)$ with numerical gradient	**M1**	Gradient must come from a differentiated expression.
	Obtain $y = -\frac{1}{2}x + \frac{31}{4}$ or equivalent of requested form	**A1**	
		4	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
7(b)	Integrate to obtain form $k_2(2x+1)^{\frac{2}{3}}$	**M1**	
	Obtain correct $9(2x+1)^{\frac{2}{3}}$ or unsimplified equivalent	**A1**	
	Use correct limits correctly to find area	**M1**	Substitute correct limits into an integrated expression. 36 – 9 minimum working required.
	Obtain 27	**A1**	**SC B1** if M1 A1 M0 scored.
		4	

Question	Answer	Marks	Guidance
8(a)	Use $\tan^2\beta = \dfrac{\sin^2\beta}{\cos^2\beta}$	**B1**	E.g. $\tan^2\beta = \dfrac{\sin^2\beta}{\cos^2\beta}$ and then replaces $\sin^2\beta$ with a^2 or $\cos^2\beta$ with $1 - a^2$.
	$\cos\beta = -\sqrt{1-a^2}$	**B1**	
	Obtain $\dfrac{a^2}{1-a^2} + 3a\sqrt{1-a^2}$	**B1**	
		3	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
8(b)	Use correct identity to obtain 3-term quadratic equation in $\sin\theta$	***M1**	
	Obtain $\sin^2\theta + 4\sin\theta + 1[=0]$	**A1**	
	Attempt to solve quadratic	**DM1**	At least as far as $\frac{-4\pm\sqrt{12}}{2}$. -15.5^o implies attempt at solving quadratic.
	Obtain 195.5	**A1**	
	Obtain 344.5	**A1FT**	Following first answer; and no others for $0° < \theta < 360°$ but must be in 4th quadrant. **SC B1** for 3.41^c and 6.01^c.
		5	

Question	Answer	Marks	Guidance
9(a)	Differentiate to obtain $5 + 12x - 9x^2$	**B1**	
	Attempt to find two critical values by solving quadratic equation or inequality	**M1**	
	Obtain values $-\frac{1}{3}$ and $\frac{5}{3}$	**A1**	**SC B1** if no method for solving the quadratic.
	Conclude $x < -\frac{1}{3}$, $x > \frac{5}{3}$	**A1FT**	**SC B1** if no method for solving the quadratic.
		4	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
9(b)	Equate first derivative to 9 and simplify to 3 term quadratic	***M1**	
	Obtain $x = \frac{2}{3}$	**A1**	**SC B1** for solving $5+12x-9x^2 = 9$ without simplifying to a 3-term quadratic.
	Use x-value and corresponding y-value to determine value of k	**DM1**	
	Obtain $k = \frac{28}{9}$	**A1**	**SC B1** for $k = \frac{28}{9}$ from solving $5+12x-9x^2 = 9$ without simplifying to a 3-term quadratic.
		4	

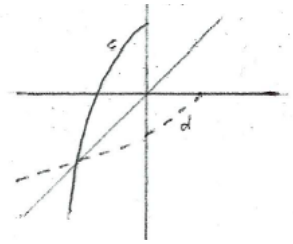
Question	Answer	Marks	Guidance
10(a)	State or imply that first 3 terms of GP are $5+d,\ 5+4d,\ 5+10d$	**B1**	
	Form equation $(5+4d)^2 = (5+d)(5+10d)$ or equivalent	**M1**	
	Obtain $d = 2.5$	**A1**	Ignore 0 as a solution. **SC B1** Obtain $d = 2.5$ and 7.5, 15, 30 by trial and improvement www.
	Alternative Method for Question 10(a):		
	State or imply that first 3 terms of GP are $5+d,\ 5+4d,\ 5+10d$	**B1**	
	$(5+d)R = 5+4d \rightarrow d = \frac{5-5R}{R-4}$, $(5+d)R^2 = 5+10d \rightarrow R^2 - 3R + 2$ [= 0]	**M1**	OE Eliminates d.
	Obtain $d = 2.5$	**A1**	
		3	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
10(b)	Use correct formula for sum of AP with their value of d	**M1**	
	Obtain or imply 7700	**A1**	
	State or imply GP is $7.5,\ 15,\ 30, ...$	**B1**	
	Use correct formula for sum of GP with their common ratio	**M1**	
	Obtain $S_{77} - G_{10} = 27.5$	**A1**	
		5	

Question	Answer	Marks	Guidance
11(a)	Obtain $b = 2$ and $c = \frac{3}{2}$	**B1**	
	Obtain $\frac{15}{2} - 2\left(x - \frac{3}{2}\right)^2$	**B1**	
	State range is $y \leqslant \frac{15}{2}$ or $f(x) \leqslant \frac{15}{2}$ with $\leqslant$ given or clearly implied (not $<$)	**B1 FT**	Following *their* value of a.
		3	
11(b)	State that reflection is in x-axis	**B1**	Accept transformations in any order.
	State or imply that translation is by $\begin{pmatrix} -\frac{3}{2} \\ \frac{15}{2} \end{pmatrix}$ or equivalent	**B1 FT**	Following *their* values of a and c in part **(a)**. Accept transformations in any order.
		2	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Guidance
11(c)	Sketch the correct graph appearing in second and third quadrants only	**B1**	
	State that each *y*-value is associated with a single *x*-value or equivalent	**B1**	Accept passes the horizontal line test. Ignore passes the vertical line test.
		2	
11(d)	Sketch the correct graph with suitable labelling to distinguish the two curves	**B1**	Appearing in third and fourth quadrants only.
	Draw the line $y = x$	**B1**	See above; no need to label the line.
	Attempt correct process for finding the inverse function	**M1**	Allowing use of $\pm$ and *y* so far.
	Obtain $\frac{3}{2} - \sqrt{\frac{15}{4} - \frac{1}{2}x}$ or equivalent	**A1**	Must involve *x* at the conclusion.
		4	

© Cambridge University Press & Assessment 2024